Cross-Modal Change Attention and Damage-Aware CutMix for Multimodal Building Damage Assessment

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***Abstract*—Multimodal building damage assessment requires reliable fusion between pre-event optical imagery and post-event synthetic aperture radar (SAR) observations, while also coping with severe damage-class imbalance and cross-event distribution shifts. This paper introduces a cross-modal change attention module and a damage-aware CutMix training strategy for all-weather building damage mapping. The method is designed as a lightweight plug-in for standard segmentation backbones and is evaluated through controlled ablation studies on the BRIGHT benchmark.**

***Index Terms*—Building damage assessment, multimodal remote sensing, optical imagery, synthetic aperture radar, cross-modal attention, data augmentation, CutMix.**

I. Introduction

RAPID building damage assessment (BDA) is a critical component of emergency response after earthquakes, floods, storms, wildfires, explosions, and armed conflicts. Building-level damage maps support search-and-rescue prioritization, loss estimation, and relief allocation, yet field surveys are often slow, dangerous, and spatially incomplete. Earth observation (EO) offers a scalable alternative by repeatedly imaging large affected areas. Consequently, optical, synthetic aperture radar (SAR), and multitemporal remote sensing have long been investigated for post-disaster damage mapping [1]–[3].

The availability of very-high-resolution (VHR) imagery and large annotated datasets has shifted BDA from handcrafted indicators toward data-driven localization and multiclass damage recognition. The xBD benchmark enabled systematic evaluation across multiple hazards and geographic regions [4]. Subsequent methods have explored joint localization and classification, multitemporal feature interaction, object-based semantic change detection, and operational generalization [5]–[8]. These studies demonstrate the potential of deep neural networks, but most large-scale BDA systems still depend on paired pre- and post-event optical observations.

Optical imagery provides intuitive spectral and textural evidence, but usable post-event acquisitions can be delayed by clouds, smoke, precipitation, or the absence of daylight. SAR actively illuminates the surface and therefore offers day-and-night, largely weather-independent observation. Multitemporal SAR studies have used intensity changes and interferometric coherence, while PolSAR studies have analyzed damage-induced changes in scattering mechanisms [2], [3], [9]–[13]. Single post-event VHR SAR studies further demonstrate the operational value of radar data when a suitable pre-event SAR acquisition is unavailable [14]. More recent work has combined optical and SAR data with statistical fusion or deep networks [15]–[18]. In particular, the BRIGHT benchmark provides open, globally distributed VHR pre-event optical and post-event SAR image pairs over 14 disaster events, together with pixel-level damage labels [19]. It creates a realistic setting in which detailed pre-disaster optical information can be paired with rapidly available post-disaster SAR observations.

This setting nevertheless poses a difficult heterogeneous change-understanding problem. Optical values describe reflected solar radiation, whereas SAR amplitudes depend on microwave scattering, viewing geometry, surface roughness, and dielectric properties. Even after geometric coregistration, corresponding optical and SAR pixels need not have similar appearance. Direct channel concatenation can therefore entangle true structural damage with modality-induced differences. Heterogeneous change-detection studies have addressed this issue through image translation, common feature spaces, and progressive modality alignment [20], [21]. Optical–SAR building extraction has likewise benefited from explicitly modeling shared and modality-specific information [22]. However, these approaches do not directly address multiclass building damage assessment on globally diverse VHR events.

A second challenge is the joint effect of class imbalance and cross-event distribution shift. Damaged and destroyed buildings occupy far fewer pixels than background and intact buildings, and their visual or scattering signatures vary with hazard mechanism, construction type, sensor, and geography. Extreme imbalance has been documented in optical BDA [23], while event-disjoint evaluations show that strong models can deteriorate markedly on unseen locations [8], [24]. Recent multi-hazard methods also emphasize the need to separate damage evidence from event-specific context [25]. Generic geometric or radiometric augmentation does not explicitly increase exposure to rare damage pixels, nor does it force damage patterns to be learned across different event backgrounds.

To address these issues, this paper proposes a Cross-Modal Change Attention (CMCA) module and a Damage-Aware CutMix (DA-CutMix) training strategy for multimodal building damage assessment. CMCA performs explicit optical-to-SAR feature comparison at the bottleneck level, using pre-event optical features as spatial queries and post-event SAR features as keys and values. DA-CutMix extends CutMix [26] to multimodal damage segmentation by sampling damage-rich donor patches from different disaster events and pasting them synchronously into the optical image, SAR image, and label map. The two components are lightweight and can be attached to multiple segmentation backbones.

The main contributions of this paper are summarized as follows.

- We propose CMCA, a lightweight cross-modal attention module that replaces simple optical/SAR concatenation with explicit change-aware feature interaction.
- We introduce DA-CutMix, an event-aware and damage-aware augmentation strategy that increases rare damage-class exposure while preserving optical–SAR–label alignment.
- We evaluate the proposed components on the BRIGHT benchmark with six representative backbones, including CNN-, transformer-, Siamese-, and Mamba-based models, and analyze both overall and per-event robustness.

II. Related Work

A. Remote Sensing for Building Damage Assessment

Traditional BDA methods derive damage cues from mono- or multitemporal EO observations. Optical approaches use spectral, textural, geometric, and object-level differences, whereas SAR approaches exploit backscatter intensity, interferometric coherence, and polarimetric scattering [1]–[3]. Multitemporal PolSAR studies associated changes in double-bounce and volume scattering with tsunami- or earthquake-induced structural damage [9], [13]. Other methods combined intensity and coherence from multiple sensors [11], although building-level coherence is constrained by spatial averaging, building size, temporal decorrelation, and acquisition geometry [12]. When pre-event SAR is unavailable, post-event PolSAR methods use scattering and texture cues to separate collapsed buildings from intact but obliquely oriented structures [10]; recent VHR SAR studies have also explored learning building-level classifiers across multiple earthquake regions [14]. At the individual-building level, Brunner *et al.* predicted an intact post-event SAR signature from pre-event VHR optical imagery and compared it with the observed response [15], while belief-fusion methods combined superpixel evidence from spaceborne optical and SAR data [16]. These studies established the value of SAR for rapid response, but also expose dependencies on sensor-specific features, ancillary building data, and acquisition configurations.

Deep learning reduces the dependence on handcrafted damage descriptors. xBD and the xView2 challenge established a large optical bitemporal benchmark with building footprints and ordinal damage labels [4]. RescueNet jointly learned building segmentation and damage assessment [5]; BDANet used multiscale fusion, cross-directional attention, and class-focused CutMix [6]; and ChangeOS cast BDA as object-based semantic change detection and evaluated transfer from natural to human-made disasters [7]. Later studies examined realistic emergency-response testing and multi-hazard adaptation, highlighting the remaining difficulty of unseen events and intermediate damage classes [8], [25].

Multimodal BDA is less mature because paired VHR optical/SAR imagery and labels are scarce. Adriano *et al.* evaluated single-modal, cross-modal, and multimodal data-availability scenarios and showed the feasibility of pre-event optical/post-event SAR mapping [17]. M3ICNet subsequently studied cross-modal and cross-resolution feature interaction for optical–SAR damage detection [18]. BRIGHT substantially broadens this direction through an open, event-diverse VHR benchmark [19]. Our work is positioned in this supervised BRIGHT setting and targets two complementary limitations: heterogeneous feature comparison and rare damage-class learning.

B. Multimodal Optical–SAR Fusion

Optical and SAR sensing mechanisms are complementary but heterogeneous. SpaceNet 6 promoted all-weather building mapping with paired VHR optical and SAR observations [27], and progressive fusion learning later extracted modality-invariant structural information for joint optical/SAR building segmentation [22]. These studies concern static building footprints rather than disaster-induced semantic change, but they show that learned fusion should preserve both shared structure and modality-specific information.

Heterogeneous change detection more directly studies how to distinguish actual scene changes from cross-sensor discrepancies. Deep homogeneous feature fusion separates semantic content from sensor style before comparison [20], whereas progressive modality alignment alternates feature alignment and change-map refinement [21]. M3ICNet performs multilevel interaction across optical and SAR branches while preserving their native resolutions [18]. These methods demonstrate that direct pixel differencing is unsuitable for heterogeneous imagery; however, image translation or global feature alignment may also suppress task-relevant damage evidence, and most existing benchmarks target binary land-cover change rather than four-class BDA.

Attention offers a content-adaptive alternative: query–key similarity controls where information is retrieved across feature sets [28]. Change-guided attention uses coarse semantic change priors to improve multiscale feature fusion [29]. Our CMCA differs in purpose and direction. It uses pre-event optical building features as queries and post-event SAR features as keys and values at the semantic bottleneck, producing an optical-conditioned SAR representation for damage reasoning rather than a generic fused context.

C. Deep Change-Detection Architectures

Deep change detection has evolved from convolutional Siamese networks toward explicit long-range spatio-temporal modeling. Early fully convolutional Siamese architectures established end-to-end feature differencing for paired remote sensing images [30]. SiamCRNN subsequently combined convolutional feature extraction with recurrent interaction to handle multisource VHR imagery [31]. Transformer-based methods address the limited receptive field of convolution by modeling nonlocal relationships. BIT compresses bitemporal features into semantic tokens and reasons about their temporal interaction [32], while DamFormer extends Siamese

Transformer encoding and multitemporal fusion to building localization and damage classification [33].

State-space models provide another route to long-range context. Mamba introduces input-dependent selective state transitions with linear sequence scaling [34], and VMamba adapts selective scanning to two-dimensional visual features [35]. ChangeMamba further tailors spatial and temporal scans to binary change detection, semantic change detection, and BDA [36]. Related remote sensing variants explore omnidirectional dense prediction and combined local–global change cues [37], [38]. These developments motivate evaluating CMCA across CNN-, Transformer-, recurrent-, and Mamba-based backbones. Unlike a backbone-specific decoder, however, CMCA is designed as a compact fusion module whose contribution can be isolated under different architectures.

D. Class Imbalance and Data Augmentation

Dense prediction commonly exhibits foreground–background and long-tailed class imbalance. Loss-level methods such as focal loss down-weight easy dominant examples [39], while long-tailed segmentation methods rebalance class-dependent feature or logit variation [40]. In BDA, the problem is compounded by the small spatial extent and event-specific appearance of damaged buildings. Wang *et al.* explicitly studied extremely imbalanced satellite BDA [23], and event-disjoint xBD experiments linked poor transfer to unequal class distributions across disasters [24]. Thus, balancing only the global loss does not ensure that rare damage patterns are observed under diverse event contexts.

Sample-mixing augmentation provides a complementary data-level solution. CutMix pastes rectangular regions between images and mixes their supervision [26]; ClassMix selects semantic regions for segmentation consistency [41]; and Copy-Paste transfers object masks to increase object and rare-category diversity [42]. For optical BDA, BDANet applied CutMix specifically to difficult damage levels and showed that class-focused mixing was more effective than indiscriminate mixing [6]. These findings motivate targeted augmentation, but they do not account for the three-way synchronization of pre-event optical data, post-event SAR data, and dense damage labels.

The proposed DA-CutMix is tailored to this multimodal and cross-event setting. Donor patches are sampled from a different disaster event and accepted only when they contain sufficient damaged or destroyed pixels. The same box is then pasted into both sensor streams and the label map. Consequently, the method increases rare-class exposure, varies the surrounding event context, and preserves optical–SAR–label correspondence.

III. Proposed Method

This section describes the proposed framework for multimodal building damage assessment. The method contains two complementary components: Cross-Modal Change Attention (CMCA), which improves optical/SAR feature fusion at the model level, and Damage-Aware CutMix (DA-CutMix), which improves damage-class exposure and cross-event robustness at the data level. The overall technical roadmap is shown in Fig. 1.

A. Problem Formulation

Given an aligned pre-event optical image $\mathbf{X}^o \in \mathbb{R}^{3\times H\times W}$ and a post-event SAR image $\mathbf{X}^s \in \mathbb{R}^{3\times H\times W}$, the objective is to predict a pixel-wise damage map

$$\hat{\mathbf{Y}} = f_\theta(\mathbf{X}^o, \mathbf{X}^s), \tag{1}$$

where $\hat{\mathbf{Y}} \in \{0,1,2,3\}^{H\times W}$. The four classes denote background, intact building, damaged building, and destroyed building, respectively. In the implementation, the SAR image is replicated to three channels and concatenated with the optical image, yielding a six-channel input. The network is trained for four-class semantic segmentation, while the binary localization label is obtained by merging the damaged and destroyed classes with the building class.

B. Overall Framework

The baseline network processes the concatenated optical/SAR input using a standard segmentation backbone. In contrast, the proposed architecture separates the two modalities in the encoder and fuses them through CMCA at the deepest semantic level. Let $\mathbf{F}_l^o$ and $\mathbf{F}_l^s$ denote the optical and SAR features at encoder level l. Shallow skip features are fused by 1×1 convolutions,

$$\mathbf{S}_l = \eta_l\left([\mathbf{F}_l^o, \mathbf{F}_l^s]\right), \tag{2}$$

where $[\cdot,\cdot]$ is channel-wise concatenation and $\eta_l(\cdot)$ is a learned projection. At the bottleneck level, CMCA estimates an explicit change-aware representation from optical-to-SAR cross-attention. The resulting change feature is concatenated with the original bottleneck features and passed to the decoder:

$$\mathbf{F}_b = \psi\left([\mathbf{F}_b^o, \mathbf{F}_b^s, \mathbf{F}_b^c]\right), \tag{3}$$

where $\mathbf{F}_b^c$ is the CMCA output and $\psi(\cdot)$ is implemented by a 1×1 convolution, batch normalization, and ReLU activation. This design keeps the decoder structure close to the original backbone while replacing simple early concatenation with explicit cross-modal comparison.

C. Cross-Modal Change Attention

Directly concatenating optical and SAR channels gives the network access to both modalities, but it does not force the model to learn where the post-event SAR response should be compared with the pre-event optical structure. CMCA addresses this limitation by using optical features as spatial queries and SAR features as keys and values.

Let $\mathbf{F}_b^o, \mathbf{F}_b^s \in \mathbb{R}^{C\times h\times w}$ be the bottleneck features of the optical and SAR branches. First, CMCA projects the optical feature into queries and the SAR feature into keys and values:

$$\begin{aligned} \mathbf{Q} &= W_q\mathbf{F}_b^o, \\ \mathbf{K} &= W_k\mathcal{S}(\mathbf{F}_b^s), \\ \mathbf{V} &= W_v\mathcal{S}(\mathbf{F}_b^s) \end{aligned} \tag{4}$$

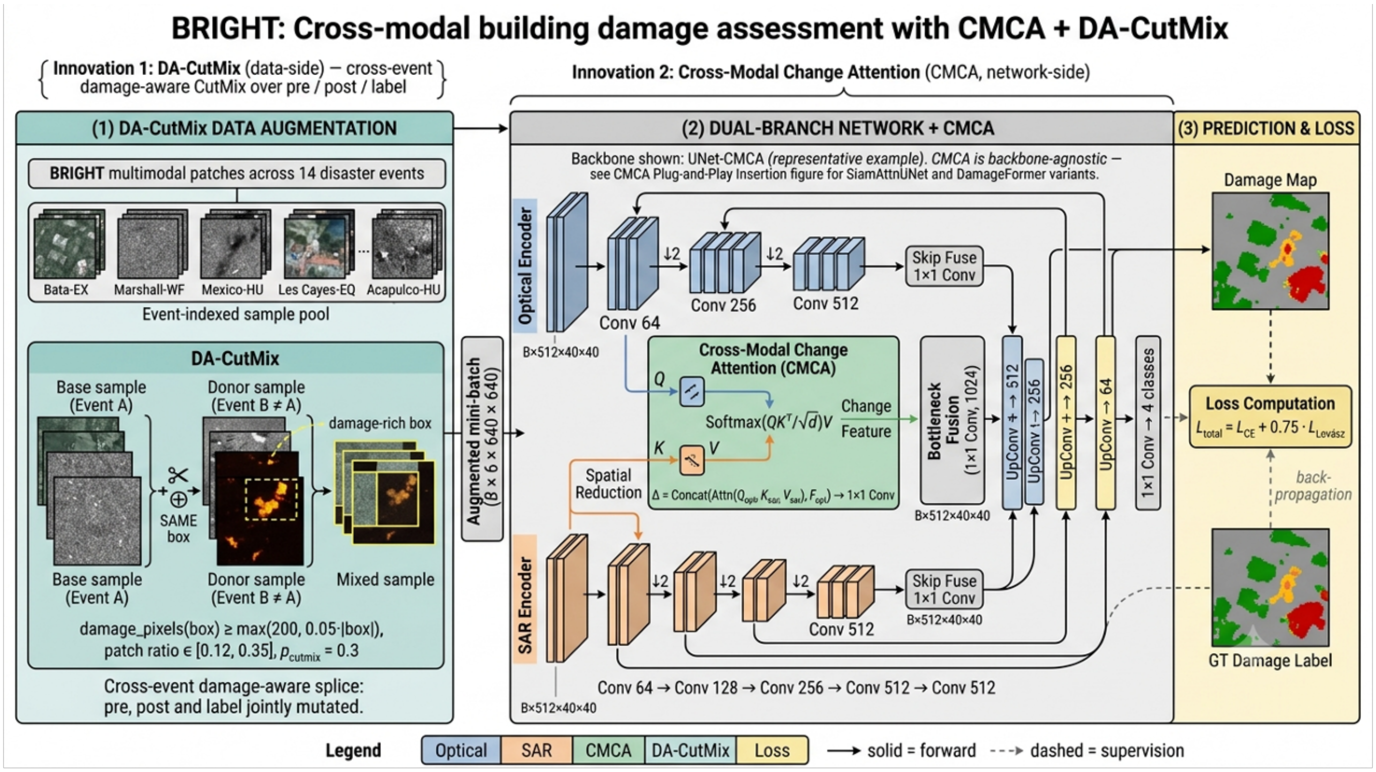

Fig. 1. Technical roadmap of the proposed multimodal building damage assessment framework. The pipeline contains architecture-level optical/SAR fusion through CMCA and data-level cross-event augmentation through DA-CutMix.

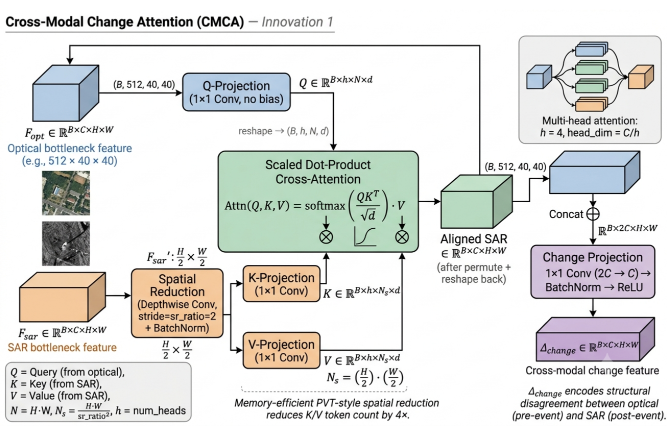

Fig. 2. Overview of the proposed Cross-Modal Change Attention module. Optical bottleneck features generate queries, while SAR bottleneck features generate keys and values after spatial reduction.

where W_q, W_k, and W_v are 1×1 convolutional projections. $\mathcal{S}(\cdot)$ is a depth-wise spatial reduction operator applied to keys and values for memory efficiency. In the current implementation, the spatial reduction ratio is set to 2.

For each attention head, the cross-modal attention map is computed as

$$\mathbf{A} = \mathrm{softmax}\left(\frac{\mathbf{Q}\mathbf{K}^\top}{\sqrt{d}}\right), \tag{5}$$

where d is the head dimension. The SAR feature aligned to optical positions is then obtained by

$$\tilde{\mathbf{F}}_b^s = \mathbf{A}\mathbf{V}. \tag{6}$$

Finally, CMCA derives a change-enhanced feature by comparing the aligned SAR representation with the optical bottleneck feature:

$$\mathbf{F}_b^c = \phi\left([\tilde{\mathbf{F}}_b^s, \mathbf{F}_b^o]\right), \tag{7}$$

where $\phi(\cdot)$ denotes a 1×1 convolution followed by batch normalization and ReLU. This formulation encourages high responses at locations where the post-event SAR structure is inconsistent with the pre-event optical building pattern, which is particularly useful for detecting severe structural damage.

The module is inserted only at the deepest encoder level. This placement reduces the quadratic cost of attention and allows the attention operation to focus on semantically rich building-level features. In the UNet implementation, both modality branches end with 512 channels, four attention heads are used, and the CMCA output is fused with the original optical and SAR bottleneck features before decoding.

D. Damage-Aware CutMix

Building damage assessment datasets are highly imbalanced: intact and background pixels dominate, while damaged and destroyed pixels are sparse. In addition, disaster events differ substantially in sensor response, urban structure, damage mechanism, and imaging geometry. To address these two issues at the data level, we introduce DA-CutMix, an event-aware and damage-aware variant of CutMix for multimodal damage segmentation. The goal is not to synthesize physically realistic disaster scenes, but to break the shortcut correlation between an event-specific background and a damage label while increasing the exposure of rare damage classes.

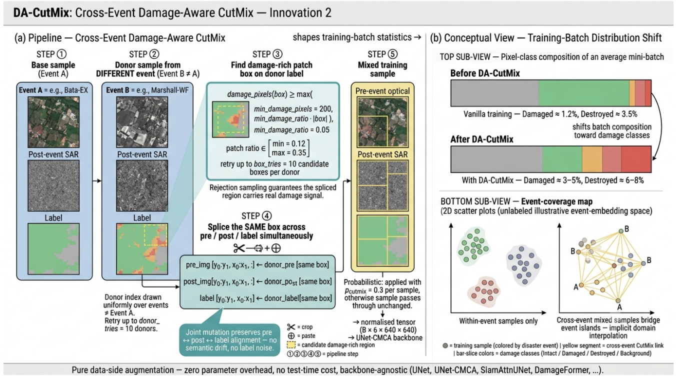

Fig. 3. Training pipeline of DA-CutMix. Donor patches are sampled from different disaster events and must contain a minimum amount of damaged or destroyed pixels before the same spatial box is pasted jointly into the optical image, SAR image, and label map.

Each training sample is represented as

$$\mathcal{T}_i = (\mathbf{X}_i^o, \mathbf{X}_i^s, \mathbf{Y}_i, e_i), \tag{8}$$

where e_i is the disaster-event identifier parsed from the sample name. Standard geometric augmentation is first applied synchronously to the optical image, SAR image, and label map. For a target sample $\mathcal{T}_i$, DA-CutMix is then attempted with probability p. A donor sample $\mathcal{T}_j$ is drawn from a different event, i.e.,, $e_j \neq e_i$, and undergoes the same type of synchronized geometric augmentation. A random rectangular box B is drawn from the donor label map. The box is accepted only when it contains enough damage-class pixels:

$$\sum_{\mathbf{u}\in B} \mathbb{I}\left[\mathbf{Y}_j(\mathbf{u}) \in \mathcal{C}_d\right] \geq \max\left(n_{\min}, \tau|B|\right), \tag{9}$$

where $\mathcal{C}_d = \{2, 3\}$ corresponds to damaged and destroyed buildings, $n_{\min}$ is the minimum number of damage-class pixels, and τ is the minimum damage-class ratio. This rejection sampling does not require all pixels in B to be damaged; the pasted patch can still contain background and intact buildings, preserving local context around the damage regions.

Let $\mathbf{M}_B$ be a binary mask that equals 1 inside the accepted box and 0 elsewhere. The mixed optical image, SAR image, and label map are generated jointly:

$$\begin{aligned}
\tilde{\mathbf{X}}^o &= \mathbf{M}_B \odot \mathbf{X}_j^o + (1 - \mathbf{M}_B) \odot \mathbf{X}_i^o, \\
\tilde{\mathbf{X}}^s &= \mathbf{M}_B \odot \mathbf{X}_j^s + (1 - \mathbf{M}_B) \odot \mathbf{X}_i^s, \\
\tilde{\mathbf{Y}} &= \mathbf{M}_B \odot \mathbf{Y}_j + (1 - \mathbf{M}_B) \odot \mathbf{Y}_i.
\end{aligned} \tag{10}$$

The same spatial box is pasted across optical, SAR, and label layers, preserving the pre-event optical, post-event SAR, and label alignment within the inserted donor patch. If no acceptable box is found after a fixed number of candidate boxes and donor samples, the original target crop is used unchanged. Unlike standard CutMix, the donor event constraint explicitly exposes the network to cross-event hybrids, while the damage-ratio constraint increases the frequency of informative damaged and destroyed regions. DA-CutMix is used only during training; inference uses the original backbone without any additional operation or test-time cost.

TABLE I
DAMAGE LABEL DEFINITIONS IN THE BRIGHT DATASET.

Category	Definition
Background (0)	Non-building pixels.
Intact (1)	Building pixels without visible structural damage or apparent disaster-related surface changes.
Damaged (2)	Buildings with partial structural damage, such as missing roof components, visible cracks, partial wall or roof collapse, or surrounding hazard effects from water, mud, burning, or volcanic material.
Destroyed (3)	Buildings that are completely collapsed, burned, severely covered by water or mud, or no longer visible at the annotated location.

E. Training Objective and Implementation Details

The main segmentation objective combines pixel-wise cross-entropy with Lovasz-Softmax loss [43]:

$$\mathcal{L} = \mathcal{L}_{\mathrm{CE}} + 0.75\,\mathcal{L}_{\mathrm{Lovasz}}. \tag{11}$$

Unless otherwise stated, no auxiliary ordinal or contrastive losses are enabled for the CMCA and DA-CutMix ablations.

The default training configuration follows the benchmark setting used in the implementation. Models are trained with AdamW [44] using a learning rate of 1×10^{-4} and a weight decay of 5×10^{-3}. The crop size is 640, the training batch size is 16, and the maximum training budget is 800,000 samples. For DA-CutMix, the damage-class set is $\mathcal{C}_d = \{2, 3\}$, $n_{\min} = 200$, $\tau = 0.05$, and up to 10 donor samples and 10 candidate boxes per donor are tested before falling back to the un-mixed target crop. The patch height and width are sampled independently between 0.12 and 0.35 of the crop size in the main runs. The mixing probability is not fixed across all backbones; it is selected from $\{0.25, 0.30\}$ on the validation setting and reported explicitly in the ablation study.

IV. EXPERIMENTS

A. Dataset and Experimental Protocol

Experiments are conducted on the BRIGHT multimodal building damage assessment benchmark. BRIGHT provides paired pre-event optical images, post-event SAR images, and pixel-level damage annotations. The annotations follow four semantic categories: background, intact building, damaged building, and destroyed building, as summarized in Table I. Following the standard supervised learning protocol in the benchmark code, the data are divided into 2917 training samples, 449 validation samples, and 877 test samples. The validation set is used for model selection, and the selected checkpoint is evaluated on the test set.

B. Evaluation Metrics

We report overall accuracy (OA), mean intersection-over-union (mIoU), localization F1 score, classification F1 score, and per-class IoU. OA is computed over all pixels. For class c, IoU is defined as

$$\mathrm{IoU}_c = \frac{\mathrm{TP}_c}{\mathrm{TP}_c + \mathrm{FP}_c + \mathrm{FN}_c}, \tag{12}$$

Fig. 4. Example DA-CutMix training sample from the BRIGHT dataset. From left to right: pre-event optical image, post-event SAR image, and damage label map. The yellow rectangle denotes the same cross-event donor region pasted into all three layers, illustrating that DA-CutMix preserves optical–SAR–label alignment while injecting a damage-rich patch.

and mIoU is the average IoU over the four semantic classes. $F1_{\mathrm{loc}}$ evaluates binary building localization by merging intact, damaged, and destroyed buildings into a single foreground class. $F1_{\mathrm{clf}}$ is the harmonic mean of the F1 scores over the three building-related classes, namely intact, damaged, and destroyed. This metric places more emphasis on balanced damage recognition than on the dominant background class.

C. Implementation Details

All methods use the same input format and data split. The pre-event optical image and post-event SAR image are concatenated into a six-channel tensor after synchronized geometric augmentation and normalization. Unless otherwise stated, models are trained for 800,000 samples with a crop size of 640, a batch size of 16, AdamW optimizer, learning rate 1×10^{-4}, and weight decay 5×10^{-3}. The loss function follows (11). For DA-CutMix, donor patches are sampled from different disaster events with damaged-class set $\{2, 3\}$, minimum damage pixels $n_{\min} = 200$, minimum damage ratio $\tau = 0.05$, and patch side length sampled from 0.12 to 0.35 of the crop size. The application probability p is selected from $\{0.25, 0.30\}$ on the validation setting; the compact DA-CutMix rows in Table II report the selected setting for each backbone.

D. Compared Variants

We evaluate the two proposed components under six representative segmentation backbones: UNet [45]; DamageFormer, the benchmark adaptation of DamFormer [33] with Swin-T encoders [46]; SiamAttnUNet [17]; SiamCRNN [31]; DeepLabV3Plus [47]; and ChangeMamba [36]. For each backbone, the baseline denotes the original model trained without CMCA or DA-CutMix. The "+CMCA" setting replaces the original optical/SAR fusion with the proposed cross-modal attention module when the backbone architecture supports a paired-branch design. The "+DA-CutMix" setting applies the proposed event-aware and damage-aware CutMix during training only. The "+CMCA+DA-CutMix" setting combines the model-level and data-level components. When a compact final comparison reports "+CMCA+DA-CutMix" without an explicit probability, it denotes the validation-selected CutMix probability rather than a single probability shared by all backbones.

E. Ablation Study

Table II summarizes the compact ablation results on the standard test split. The table is organized by backbone so that the baseline, DA-CutMix-only, CMCA-only, and combined CMCA+DA-CutMix settings can be compared directly. For rows involving DA-CutMix, the reported result uses the validation-selected probability from $\{0.25, 0.30\}$; the probability is omitted from the method name to keep the main table focused on component-level comparisons.

For UNet, CMCA improves mIoU from 64.74% to 65.39%, with a notable gain of 1.82 percentage points on the damaged class. This indicates that explicit optical-to-SAR feature interaction helps the network recover damage-sensitive structural cues that are weakly exploited by direct channel concatenation. Standalone DA-CutMix does not improve the UNet baseline in terms of mIoU, but it becomes effective when combined with CMCA. The best UNet setting, CMCA+DA-CutMix with $p = 0.30$, reaches 66.16% mIoU, improving the baseline by 1.42 percentage points. It also gives the highest per-event average mIoU of 51.59 among the UNet variants. Therefore, the validation-selected CMCA+DA-CutMix setting for UNet uses $p = 0.30$.

DeepLabV3Plus provides another convolutional encoder–decoder baseline and shows a clear benefit from the proposed components. Standalone DA-CutMix with $p = 0.25$ improves mIoU from 63.85% to 65.25% and gives the highest $F1_{\mathrm{clf}}$ in this group. CMCA alone improves mIoU to 64.83%, while the combined CMCA+DA-CutMix setting with $p = 0.25$ achieves the best mIoU of 65.79%, corresponding to a 1.94-point gain over the baseline. This setting also gives the best $F1_{\mathrm{loc}}$, OA, background IoU, intact IoU, destroyed-class IoU, and per-event average mIoU of 49.08 for DeepLabV3Plus. The $p = 0.30$ combined setting instead gives the best damaged-class IoU, increasing it from 37.06% to 40.20%, which indicates that a larger mixing probability may favor harder damage pixels while slightly reducing destroyed-class

TABLE II
OVERALL ABLATION RESULTS ON THE BRIGHT BENCHMARK. VALUES ARE REPORTED IN PERCENTAGE.

Method	$\mathrm{F1}_{\mathrm{loc}}$	$\mathrm{F1}_{\mathrm{clf}}$	OA	mIoU	Background	Intact	Damaged	Destroyed
UNet	88.32	70.91	95.52	64.74	96.28	71.62	37.48	53.59
UNet + DA-CutMix	88.07	70.89	95.48	64.46	96.22	71.43	38.05	52.15
UNet + CMCA	88.68	**72.65**	95.59	65.39	96.29	72.79	**39.30**	53.20
UNet + CMCA + DA-CutMix	**89.18**	72.43	**95.81**	**66.16**	**96.50**	**73.58**	39.08	**55.49**
DamageFormer	89.88	72.61	96.00	66.84	96.70	74.62	39.97	56.06
DamageFormer + DA-CutMix	**89.94**	73.05	**96.09**	**67.20**	**96.77**	**74.94**	**41.40**	55.70
DamageFormer + CMCA	89.90	71.96	95.97	66.55	96.72	74.35	38.61	56.53
DamageFormer + CMCA + DA-CutMix	89.91	**73.07**	96.00	67.14	96.71	74.54	40.17	**57.12**
SiamAttnUNet	88.26	71.79	95.51	65.03	96.21	72.06	38.24	53.60
SiamAttnUNet + DA-CutMix	87.92	71.85	95.49	65.28	96.11	71.95	37.45	55.62
SiamAttnUNet + CMCA	88.08	72.15	95.49	65.05	96.19	71.67	**39.50**	52.84
SiamAttnUNet + CMCA + DA-CutMix	**88.44**	**72.67**	**95.67**	**66.28**	**96.33**	**72.55**	38.94	**57.30**
SiamCRNN	**89.11**	72.12	95.83	66.13	**96.49**	73.63	40.22	54.17
SiamCRNN + DA-CutMix	89.05	72.15	**95.85**	66.36	**96.49**	**73.67**	39.20	**56.09**
SiamCRNN + CMCA	88.85	72.85	95.75	66.25	96.42	73.12	**41.26**	54.21
SiamCRNN + CMCA + DA-CutMix	89.04	**73.13**	95.75	**66.52**	96.41	73.49	40.10	56.06
DeepLabV3Plus	87.36	69.91	95.39	63.85	96.05	70.82	37.06	51.45
DeepLabV3Plus + DA-CutMix	87.68	**71.77**	95.53	65.25	96.10	71.81	**39.80**	53.30
DeepLabV3Plus + CMCA	87.71	69.76	95.56	64.83	96.23	71.14	38.18	53.76
DeepLabV3Plus + CMCA + DA-CutMix	**88.79**	71.22	**95.80**	**65.79**	**96.44**	**73.32**	39.26	**54.12**
ChangeMamba	89.95	73.18	95.83	66.73	96.42	73.66	40.74	56.09
ChangeMamba + DA-CutMix	89.75	73.54	95.89	67.04	96.47	73.88	42.21	55.60
ChangeMamba + CMCA	89.93	**74.41**	95.87	67.32	96.45	73.95	**42.55**	56.34
ChangeMamba + CMCA + DA-CutMix	**90.12**	74.31	**95.92**	**67.43**	**96.54**	**74.12**	41.82	**57.25**

consistency. We therefore select $p = 0.25$ for the compact DeepLabV3Plus+CMCA+DA-CutMix setting.

The stronger backbones reveal a more nuanced pattern. For DamageFormer, standalone DA-CutMix with $p = 0.30$ gives the best overall mIoU in this group, increasing mIoU from 66.84% to 67.20% and damaged-class IoU from 39.97% to 41.40%. However, the per-event averages show that the baseline and $p = 0.25$ setting are more stable across events, with average mIoU values of 52.04 and 52.02, respectively, compared with 51.68 for $p = 0.30$. The combined DamageFormer setting with $p = 0.30$ does not exceed the standalone DA-CutMix mIoU, but it achieves the best destroyed-class IoU in the group, improving it from 56.06% to 57.12%, and also gives the highest $F1_{\mathrm{clf}}$ of 73.07%. This suggests that the two components may emphasize different damage categories depending on the backbone; for DamageFormer, $p = 0.30$ is preferable for overall mIoU, while $p = 0.25$ or the baseline behavior is more conservative for event-level robustness.

For SiamAttnUNet, CMCA alone raises $F1_{\mathrm{clf}}$ from 71.79% to 72.15% and damaged-class IoU from 38.24% to 39.50%, while leaving overall mIoU nearly unchanged at 65.05%. The combined CMCA+DA-CutMix setting with $p = 0.25$ achieves the best overall mIoU of 66.28%, a 1.25-point gain over the baseline, and improves $F1_{\mathrm{clf}}$ to 72.67% and destroyed-class IoU to 57.30%. Across the 13 available per-event results, this combined setting averages 50.73% mIoU, improves the baseline average by 1.90 points, and obtains the best result on five events. Standalone DA-CutMix with $p = 0.30$ has the highest per-event average of 51.31%, indicating that the setting with the best global test mIoU need not maximize the unweighted event average. We therefore use $p = 0.25$ for the compact SiamAttnUNet+CMCA+DA-CutMix row. For SiamCRNN, the baseline is already competitive, but the proposed components still improve damage recognition. CMCA alone raises damaged-class IoU from 40.22% to 41.26%, while CMCA+DA-CutMix with $p = 0.25$ gives the best mIoU, $F1_{\mathrm{clf}}$, and per-event average mIoU, reaching 66.52%, 73.13%, and 51.11, respectively. The $p = 0.30$ combined setting gives the best destroyed-class IoU for SiamCRNN, but the validation-selected compact setting uses $p = 0.25$.

ChangeMamba further tests whether the proposed components remain useful for a Mamba-based long-range context model. The baseline reaches 66.73% mIoU, while DA-CutMix with $p = 0.30$ improves it to 67.04%. CMCA alone gives a stronger gain, increasing mIoU to 67.32% and damaged-class IoU from 40.74% to 42.55%. The combined ChangeMamba+CMCA+DA-CutMix setting with $p = 0.30$ achieves the best mIoU of 67.43% and the best destroyed-class IoU of 57.25%, suggesting that explicit cross-modal comparison remains complementary to the global context modeling capacity of ChangeMamba.

F. Per-Event Robustness

We further inspect per-event mIoU to assess whether the gains are concentrated on a small number of events or distributed across disaster scenarios. In the UNet ablation, the CMCA+DA-CutMix setting with $p = 0.30$ obtains the best per-event mIoU on 10 out of the 15 evaluated events, and its average per-event gain over the UNet baseline is 3.50 percentage points. The $p = 0.25$ combined setting is the best on four additional events, while standalone CMCA is best on one event. These results suggest that the proposed components are especially useful for improving event-level robustness rather than only optimizing the global pixel distribution.

The per-event behavior also shows that the optimal mixing probability is backbone-dependent. DamageFormer illustrates the clearest trade-off: standalone DA-CutMix with $p = 0.30$ gives the best global mIoU, whereas the baseline and $p = 0.25$

variants are more stable across events. SiamAttnUNet shows a related but distinct trade-off: the compact combined row uses $p = 0.25$ and gives the highest global mIoU, whereas standalone DA-CutMix with $p = 0.30$ gives the highest unweighted event average. For DeepLabV3Plus, CMCA+DA-CutMix with $p = 0.25$ wins on seven out of the 15 evaluated events and yields an average per-event gain of 2.29 percentage points over the baseline. For ChangeMamba, the combined CMCA+DA-CutMix setting with $p = 0.30$ gives the highest reported per-event average mIoU of 51.82 and obtains the best result on five events. Overall, the final compact "+CMCA+DA-CutMix" setting should not be interpreted as a fixed $p = 0.30$ configuration. Instead, it denotes a validation-selected probability: $p = 0.30$ for UNet+CMCA and ChangeMamba+CMCA, $p = 0.25$ for SiamAttnUNet+CMCA, SiamCRNN+CMCA, and DeepLabV3Plus+CMCA, and a separate overall/event-level trade-off for DamageFormer.

V. Conclusion

This paper presented CMCA and DA-CutMix for multi-modal building damage assessment with pre-event optical and post-event SAR imagery. CMCA introduces explicit optical-to-SAR feature interaction at the bottleneck level, while DA-CutMix improves rare damage-class exposure by pasting damage-rich cross-event patches synchronously across the optical image, SAR image, and label map. Experiments on the BRIGHT benchmark show that the proposed components are effective across multiple segmentation backbones, including CNN-, Siamese-, transformer-, and Mamba-based models.

The results indicate that the two components are complementary but backbone-dependent. CMCA consistently improves damage-sensitive feature fusion, whereas the best DA-CutMix probability varies across architectures and events. This suggests that all-weather BDA benefits from treating optical/SAR fusion and damage-aware data augmentation as coupled design choices. Future work will investigate more adaptive patch selection, uncertainty-aware event generalization, and stronger exploitation of SAR-specific scattering priors.

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